

Figure 1: Firewall

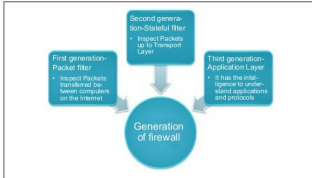


Figure 2: Generation of firewall

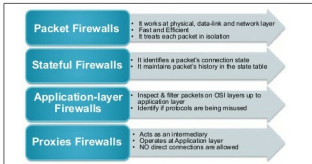


Figure 3: Types of firewalls

iptables architecture comprises groups of network packets, processing rules into tables and chains for processing the rules. Rules consist of matches to determine which packet the rule will apply to and the targets. They operate at the OSI layer, i.e., the network layer.

For more details, you can visit <https://www.youtube.com/watch?v=QF2vEvqHgU>

To verify the status of iptables, execute the following command:

```
service iptables status
```

To start and stop the iptables service, use the following command:

```
service iptables start / stop
```

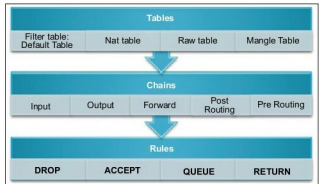


Figure 4: iptables architecture

```
root@localhost:~#
root@localhost:~# service iptables status
iptables v1.4.21: loaded successfully
Chain INPUT (policy ACCEPT)
num target prot opt source destination
0 ACCEPT tcp -- 0.0.0.0/0 0.0.0.0/0 state RELATED,ESTABLISHED
1 ACCEPT tcp -- 0.0.0.0/0 0.0.0.0/0
2 ACCEPT tcp -- 0.0.0.0/0 0.0.0.0/0
3 REJECT all -- 0.0.0.0/0 0.0.0.0/0 state NEW tcp dpt:22 reject-with tcp-host-prohibited
Chain FORWARD (policy ACCEPT)
num target prot opt source destination
1 REJECT all -- 0.0.0.0/0 0.0.0.0/0 reject-with icmp-host-prohibited
Chain OUTPUT (policy ACCEPT)
num target prot opt source destination
```

Figure 5: The status of iptables

To open the iptables file, execute the following command is:

```
gedit /etc/sysconfig/iptables
```

Syntax for executing iptable command:

```
iptables -A chain firewall-rule
```

iptables parameters	Details
-A	Add/append rule
-p	To indicate the protocol for the rule (tcp, udp, icmp, etc)
-s	To indicate the source of the packet (ip address, network address or hostname)
-d	To indicate the destination of the packet
-j	'Jump to target' indicates what should happen to the packet that matches this firewall rule
-i	'Input interface' indicates the interface through which the incoming packets are coming through the INPUT, FORWARD and PREROUTING chain
-o	'output interface' indicates the interface through which the outgoing packets are sent through the INPUT, FORWARD and PREROUTING chain
-sport	Source port for -p tcp, or -p udp
-source-port	
-dport	Destination port for -p tcp, or -p udp
-destination-port	